

Development of intelligent systems (RInS)

Task 2: Industry 5.0

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INDUSTRY 5.0



2026 Final task – Industry 5.0

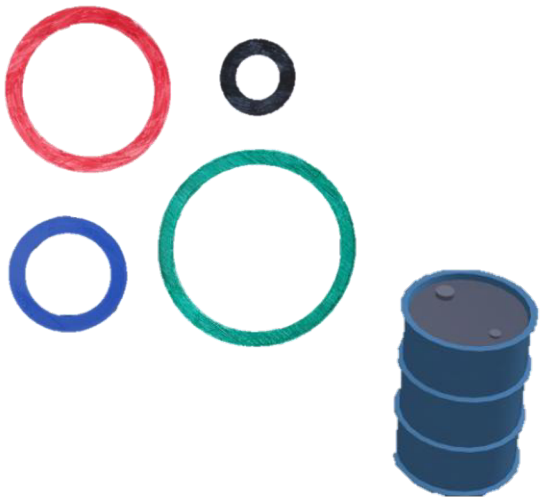


- Setup:
 - „A factory plant“ scene. Paths, working cells, conveyor belts, obstacles.
 - Several working cells, containing objects to be inspected, barrels, rings.
 - Several workers, CTO.
- Goal:
 - Help human workers by executing visual inspection and counting tasks, while navigating safely and communicating clearly.
- Task:
 - Navigate safely through the factory while staying on permitted paths.
 - Detect and approach workers to receive task instructions.
 - Identify and localise the requested working cell.
 - Capture high-resolution images of the working cell and its objects.
 - Execute the prescribed tasks (defect detection, counting, inspection).
 - Generate an inspection report and present it to the CTO.

Simulated scene

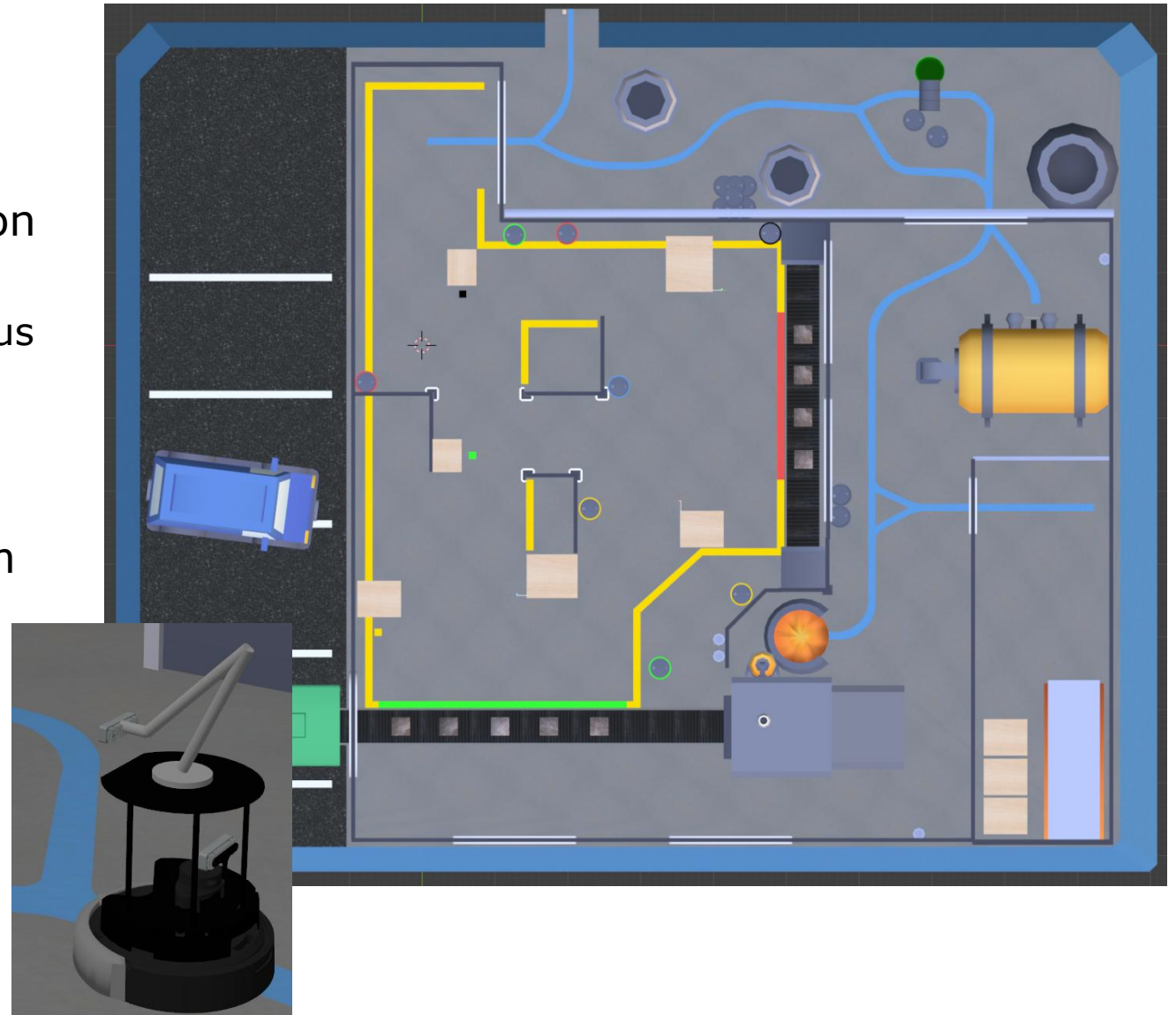


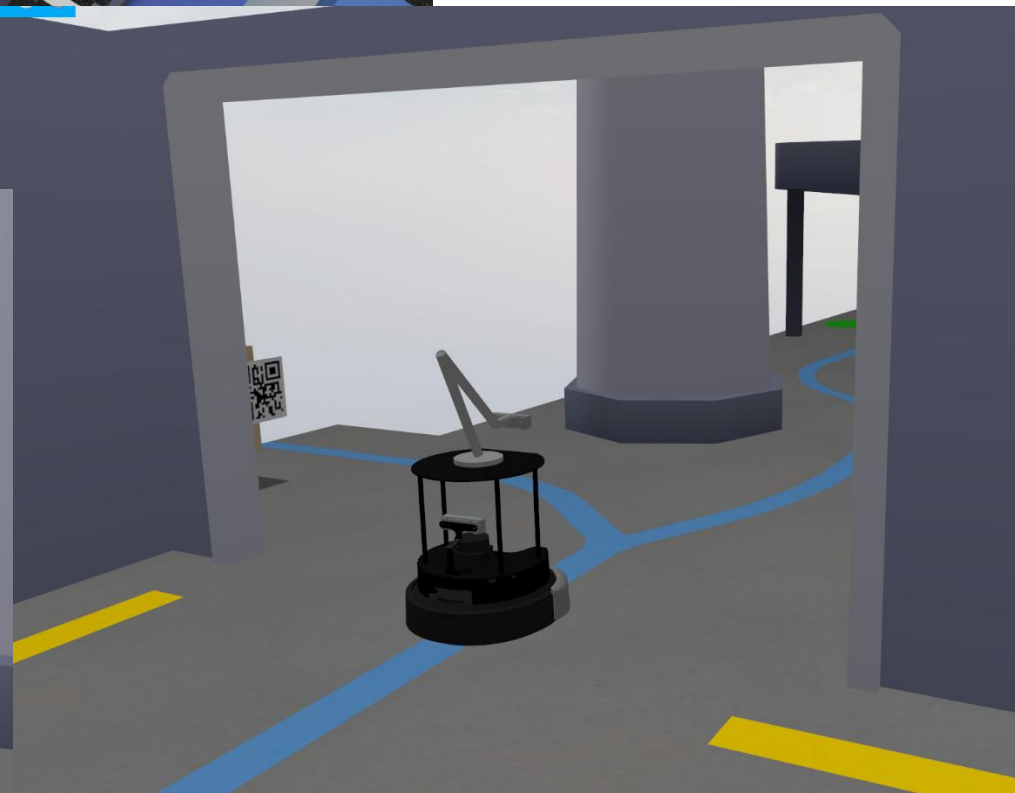
Elements in the scene



Navigation

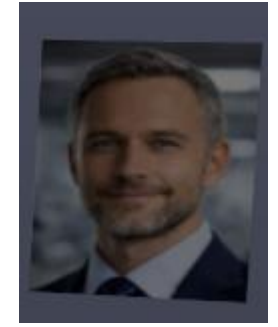
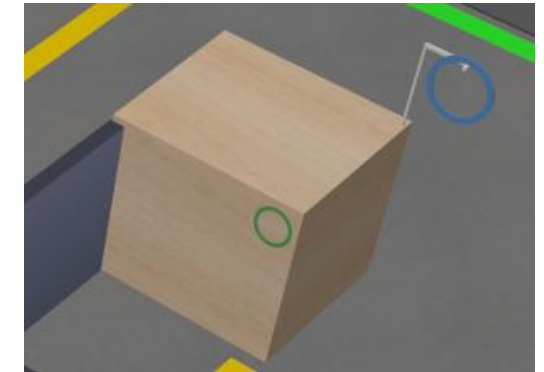
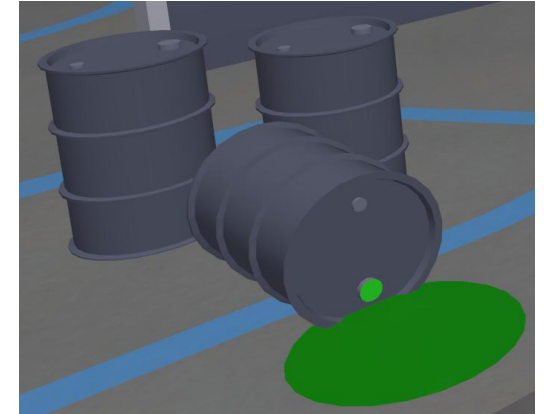
- Explore the factory plant
 - Avoid all obstacles
- In the first room
 - Use standard (goal-based) navigation
 - by using a predetermined path or
 - preferably implement the autonomous exploration of space
 - Do not cross the yellow lines
 - You can use the range sensor
 - You can set the goal at the exit from this room
- In the second room
 - Follow the blue lines
 - Use visual surveillance
 - Use the second camera pointed downwards to see what is there
 - Find CTO at the end of the blue line





Perception

- Line detection
 - To detect forbidden zones
 - For line following
 - To determine the correct working cell
- Inspect the barrels
 - Locate all barrels (in the first room only)
 - Recognise barrel colour
 - Determine orientation (horizontal or vertical)
 - Check whether the horizontal barrel is leaking
 - Approach horizontal barrels and issue a warning if a spill is detected
- Count the rings
 - Detect all (3D) rings (discard 2D printed circles)
 - Recognise ring colour
- Find all persons
 - Recognise the person's name and gender
 - Approach and engage in dialogue
- The number of any element present is not known in advance
- Colours: red, green, blue, yellow, purple, orange, brown.
- Anomaly detection



Perception and dialogue

- Engage in a dialogue with people to determine which task to perform.
 - Approach a person
 - Recognise the person's name
 - The photos and names of all persons will be given in advance
 - Recognise the person's gender
 - Ask the person what task to perform
 - Barrels inspection (in the first room only)
 - Count the rings (in the first room only)
 - Anomaly detection in red|green working cell
 - Nothing
 - Men always tell the task directly
 - Women might reconsider their choice
 - The robot has to ask them twice or more
 - When the same task is named twice, this is the task you are looking for
 - The person can also confirm the selected task with „yes.“

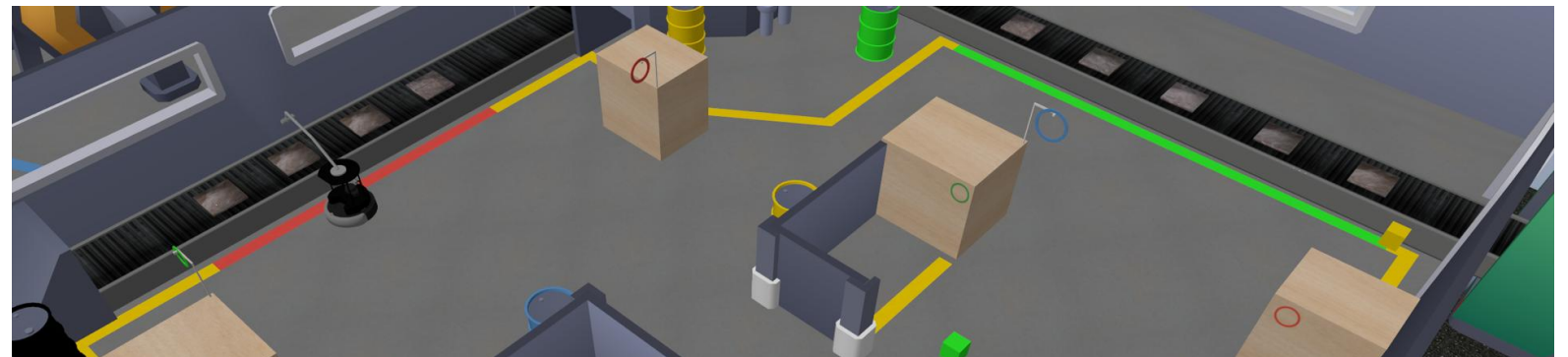
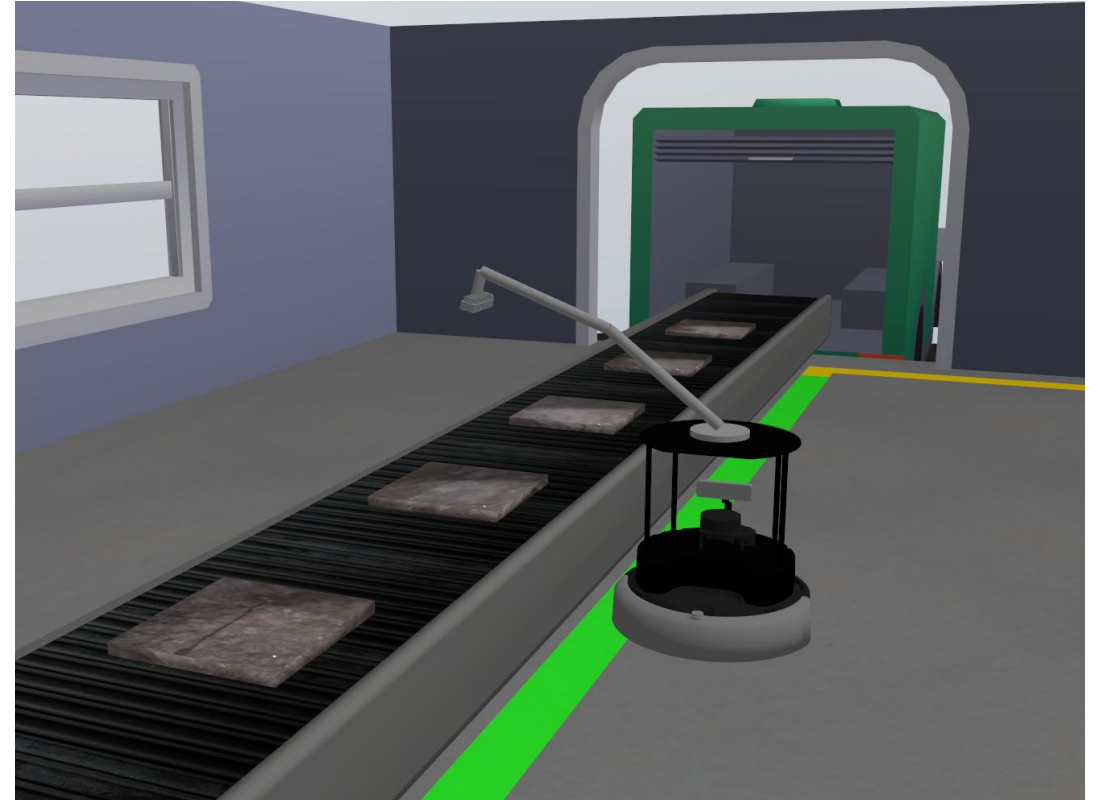
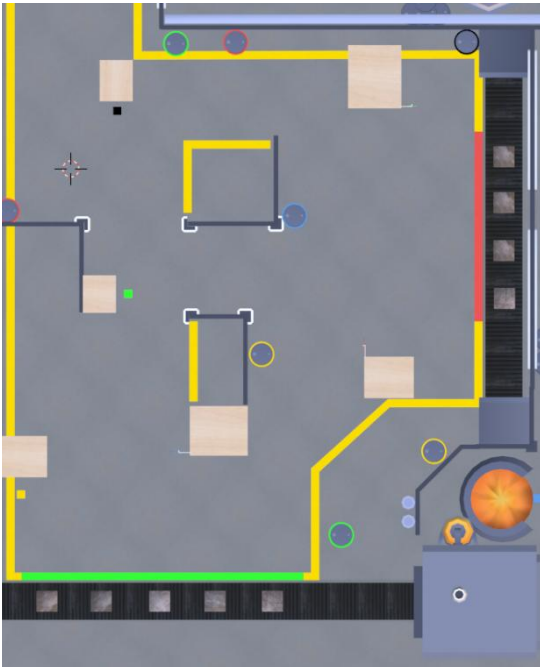


Sample dialogues

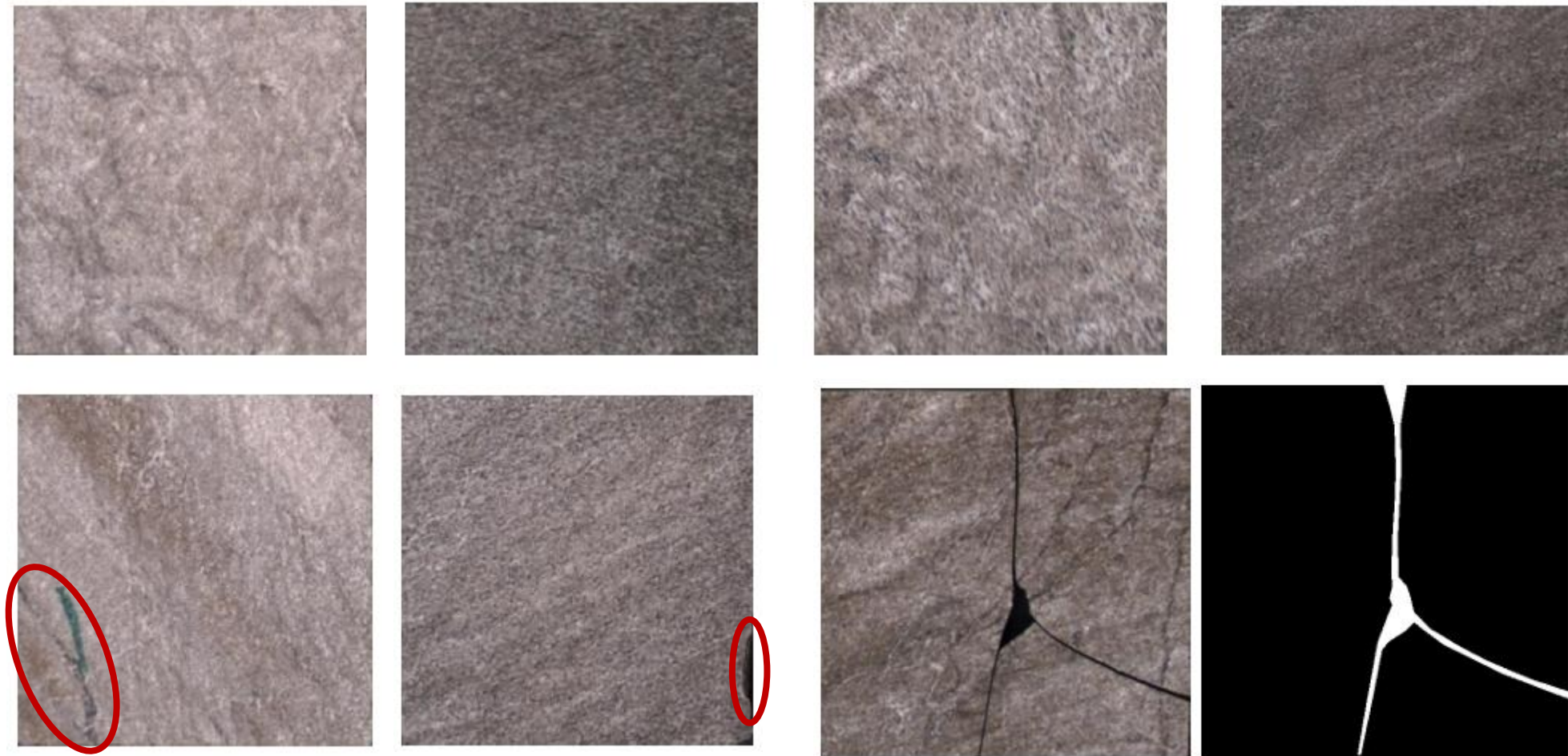
- R: „Hi man, which task should I perform?“
- M1: „Detect anomalies in the green cell.“
- R: „OK. I will detect anomalies in the green cell.“
- R: „Hi man, which task should I perform?“
- M2: „Inspect the barrels.“
- R: „OK. I will inspect the barrels.“
- R: „Alert! Alert! This barrel is leaking!“
- R: „Hi woman, which task should I perform?“
- W1: „Count the rings.“
- R: „Are you sure?“
- W1: „Well, maybe you should inspect the barrels?“
- R: „OK, the barrels then. Are you sure?“
- W1: „Ah, not really. The count the rings.“
- R: „OK. I will count the rings.“
- R: „Hi woman, which task should I perform?“
- W2: „Detect anomalies in the red cell.“
- R: „Are you sure?“
- W2: „Yes, I`m sure.“
- R: „OK. I will inspect anomalies in the red cell.“
- R: „Hi woman, which task should I perform?“
- W3: „Inspect anomalies in the red cell.“
- R: „Are you sure?“
- W3: „Well, inspect anomalies in the green cell instead.“
- R: „OK, the green cell then. Are you sure?“
- W3: „Ah, not really. Just count the rings.“
- R: „Are you sure now?“
- W3: „Yes.“
- R: „OK. I will count the rings.“

Anomaly detection

- Locate the correct working cell
 - Red or green
- Extend the camera above the objects
- Move along the working cell
- Take images of all objects (tiles)
- Detect anomalies (defects)



Anomaly detection



Anomaly detection

- Create/train a surface anomaly detector
 - Based on PCA, AE, VAE, normalising flow, diffusion models, discriminative, reconstructive methods, etc.
 - Supervised or unsupervised learning
 - Training a couple of test images will be given in advance
 - Train an anomaly detection model
 - Two levels of difficulty:
 - Severe degradations (cracks, etc.)
 - Mild degradations (more difficult to detect)
 - Take a number of training images and augment them
 - In Gazebo, in simulation, as you want
 - Taken from the frontal view or from side views as well
 - Vary the illumination conditions etc.
 - Use or design a method for anomaly detection
 - Train the model
- Evaluate the anomaly detection model
 - Create also test images
 - Evaluate the model under different environments
- The robot should detect and segment the anomaly
- Can be done (almost) completely outside ROS and Gazebo



Inspection report in pdf file

Inspection report

Date: 22.4.2026

Robot: R2D2

Task: Ring Counting

Requested by: Tom

Results:

Total number of rings detected: 5

Detected colors:

Red: 2

Blue: 1

Green: 2

Task: Barrel inspection

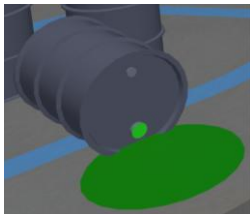
Requested by: Ana

Results:

Total number of barrels inspected: 3

Barrel ID	Colour	Orientation	Leak detected
1	Red	Vertical	No
2	Green	Horizontal	No
3	Blue	Horizontal	Yes

ID 3:



Task: Anomaly detection

Requested by: Janez

Results:

Total number of tiles inspected: 3

OK: 2

NOK: 1

Tile ID	Status
1	OK
2	OK
3	NOK

ID 3:



Shortcuts

- You may not implement all the functionalities (for a lower grade)
- You may not implement the autonomous exploration of space
 - And can use fixed goals instead
 - The first and the exit point in the first room can always be hardcoded
- The robot can skip the dialogue with a person
 - And read the task requested in the QR code next to the person
- The robot does not need to classify the faces according to gender
 - You assume that all the persons answer the right answer right away
- The robot does not need to recognise persons
 - It just enumerates the persons
- The robot does not need to avoid crossing yellow line
 - It can navigate cross the as well
- The robot does not need to follow the blue line
 - You can set a goal in front of the CTO in the second room
- The robot can skip one of the tasks
 - Either ring counting, barrel inspection or anomaly detection
- The robot can skip anomaly segmentation
 - It can report defects on the level of images only
- The robot does not need to create the inspection catalogue
 - It just reports the results during operation

Demonstration

- Demonstrate what is going on in the robot
- Visualize in RViz:
 - Locations of detected faces, rings, barrels, tiles
 - Recognised colours, faces
 - Navigation goals, path plans
 - Current sensor readings (images, Lidar)
 - Show live stream from both cameras
- Show the dialogue in a separate window
- Show the reasoning process
- Show also the current environment in Gazebo
- Information in the console does not suffice!
- Use also sound to demonstrate what is going on

R: „Hi woman, which task should I perform?“

W2: „Detect anomalies in the red cell.“

R: „Are you sure?“

W2: „Yes, I'm sure.“

R: „OK. I will inspect anomalies in the red cell.“



Tasks

- System setup
 - Running ROS
 - Tele-operating TurtleBot
- Autonomous navigation
 - Autonomous control of the mobile platform
 - Acquiring images and 3D information
 - Simultaneous mapping and localization (SLAM)
 - Path planning, obstacle avoidance
 - Advanced fine manoeuvring and basic mobile manipulation
 - Intelligent navigation and exploration of space
- Advanced perception and cognitive capabilities
 - Detection of faces, 3D rings, cylinders, and objects
 - Recognition of faces and ring and cylinder colours
 - Line detection
 - Object detection and counting
 - Defect detection
 - Speech synthesis, speech recognition, dialogue processing (reading QR codes)
 - Belief maintenance, reasoning, planning

Task 1

Task 2

Evaluation protocol

- The evaluation course will be set up in advance
 - The main setup will not change
- The teams will be allowed to build the map in advance
 - The yellow lines will not be moved, they can be considered to alter the map
- The faces, rings, cylinders and tiles will be positioned on the day of the evaluation
 - The size and colours of the rings and cylinders are known in advance
- The robot has to operate completely autonomously
 - only the initial positioning is allowed
 - Also the exit from the first room can be hardcoded
 - (and the optional answering by typing the text)
- The robot starts at the starting position in the first room
- Every team will have allocated 15-20 minutes to show the performance of the robot
- Show the performance of your robot on the lab computer (not on a slow laptop)
- The evaluation will take place in the last week of the May

Tasks

- System setup
 - Running ROS For 6
 - Tele-operating TurtleBot For + max. 2
- Autonomous navigation For + max. 2
 - Autonomous control of the mobile platform
 - Acquiring images and 3D information
 - Simultaneous mapping and localization (SLAM)
 - Path planning, obstacle avoidance
 - Advanced fine manoeuvring and basic mobile manipulation
 - Intelligent navigation and exploration of space
- Advanced perception and cognitive capabilities
 - Detection of faces, 3D rings, cylinders, and objects
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Grading

- Must do:
 - Goal-based navigation (1 pt)
 - Face detection (2 pts)
 - Ring detection (2 pts)
 - Cylinder detection (2 pts)
 - Colour recognition (1 pt)
 - Approaching faces (1 pt)
 - Speech synthesis (1 pt)
- Should do:
 - Face recognition (1 pt)
 - Gender recognition (1 pt)
 - Auton. space exploration (1 pt)
 - Not crossing yellow line (2 pts)
 - Follow blue line (2 pts)
 - Correct cell detection (1 pt)
 - Tile detection (1 pt)
 - Defect detection (3 pt)
 - Dialogue with ASR (2 pts)
 - Creating the inspection report (1 pt)
- Performance evaluation
 - Navigation (1 pt)
 - Reasoning (1 pt)
 - Visualisation (1 pt)
 - Robustness (2 pts)
 - Relative speed (2 pts)
 - Overall impression (3 pts)
- Points:
 - Must do: 10
 - Should do: 15
 - Performance: 10
 - Total: 35

Task 2 goals

- The main goals of the third task and evaluation are:
 - to navigate the robot around
 - to detect faces in 2D
 - to detect and count objects (rings and cylinders) in 3D
 - to learn and recognize colours
 - to classify faces according to gender
 - to recognise faces
 - to detect and follow lines
 - to detect defects on the tiles
 - to do simple reasoning
 - to do simple dialogue processing
 - to plan adequate actions
 - to fine manoeuvre the robot
 - to do simple mobile manipulation
 - **to integrate all functionalities into a coherent system**